

2110531 - Data Science and Data Engineering Tools

Overview

Asst. Prof. Natawut Nupairoj, Ph.D.

Department of Computer Engineering, Chulalongkorn University

natawut.n@chula.ac.th

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What You Will Learn from this Class

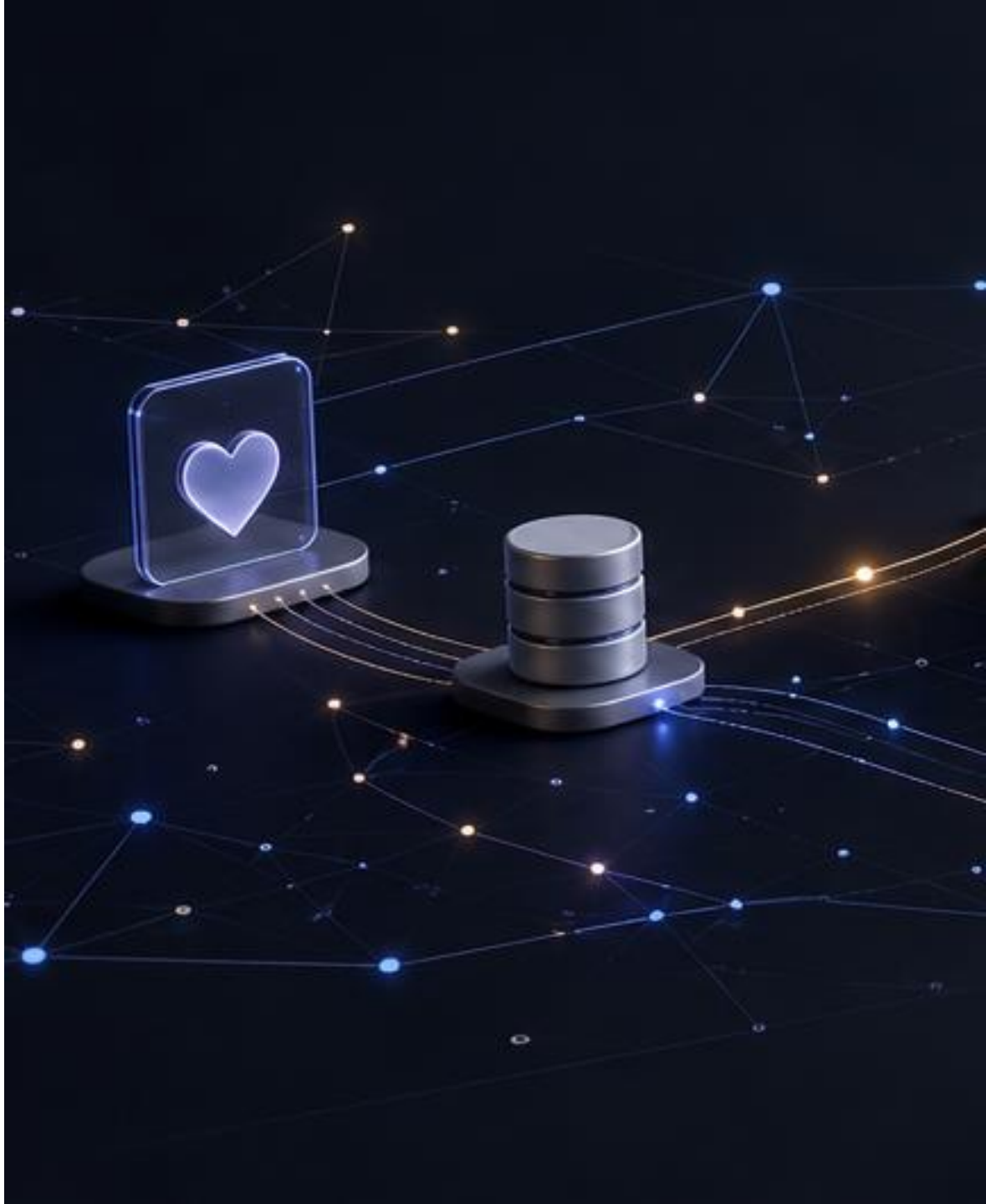
- Understand the role of data engineer and necessary skills
- Explore data pipeline across its journey—from source systems to end users
- Explain the core concepts behind big data platforms and data lakehouse architectures
- Understand data pipeline components: extraction, ingestion, processing, and orchestration

Schedule

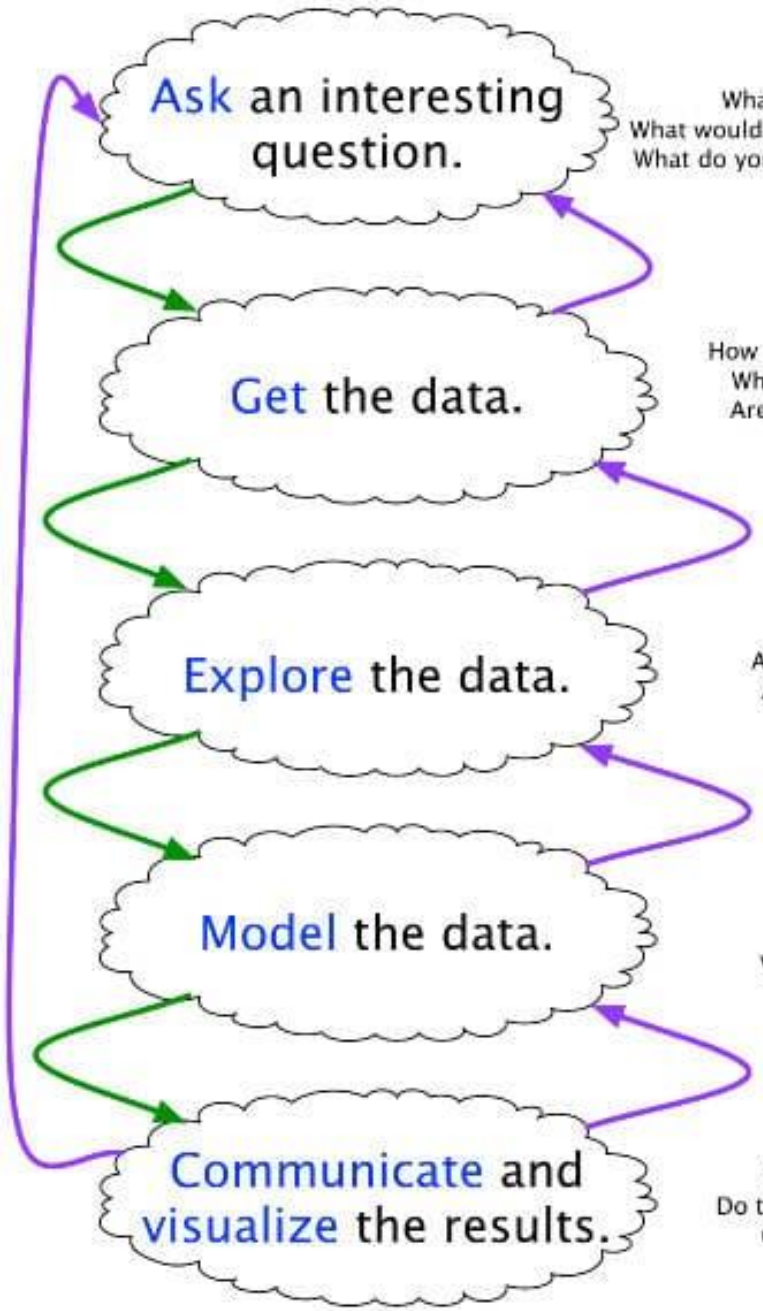
Section-1	Section-5	Topics	Note
6 Oct 2026	10 Oct 2026	Intro to Data Engineering Data Lakehouse	Iceberg / Ducklake
17 Oct 2026		Data Scraping Data Ingestion	BS4 / Selenium / Kafka
20 Oct 2026	24 Oct 2026	Data Processing	Spark
27 Oct 2026	31 Oct 2026	Pipeline Orchestration	Mlflow / Prefect
28 Nov 2026		Final Exam	

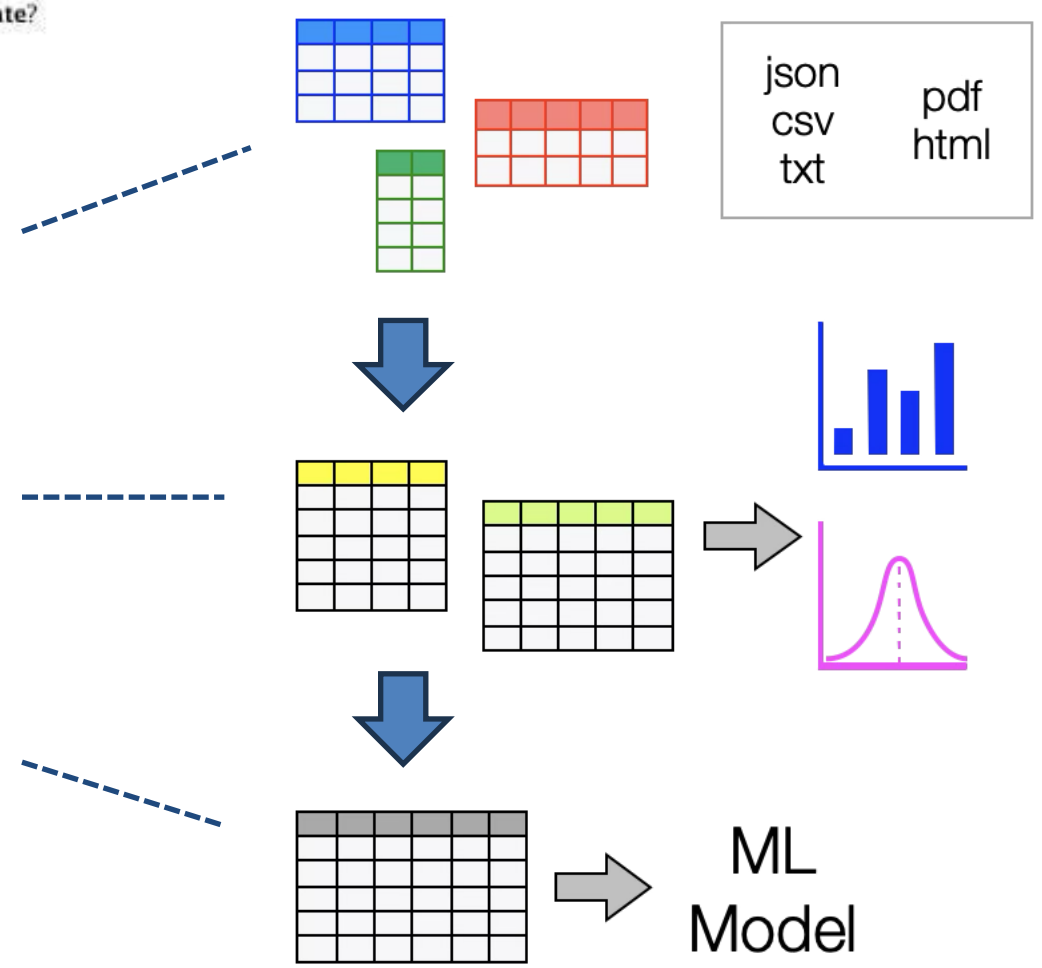
Data Science and Data Engineering Tools

Introduction to Data Engineering



The Data Science Process





Joe Blitzstein and Hanspeter Pfister, created for the Harvard data science course http://cs109.org/.

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What is Data Engineering? (Wikipedia)

- Data engineering refers to the building of systems to enable the collection and usage of data
- This data is usually used to enable subsequent analysis and data science, which often involves machine learning
- Making the data usable usually involves substantial compute and storage, as well as data processing

The Data Engineer Job Market in 2024 [Research on 1,000 Job Postings]

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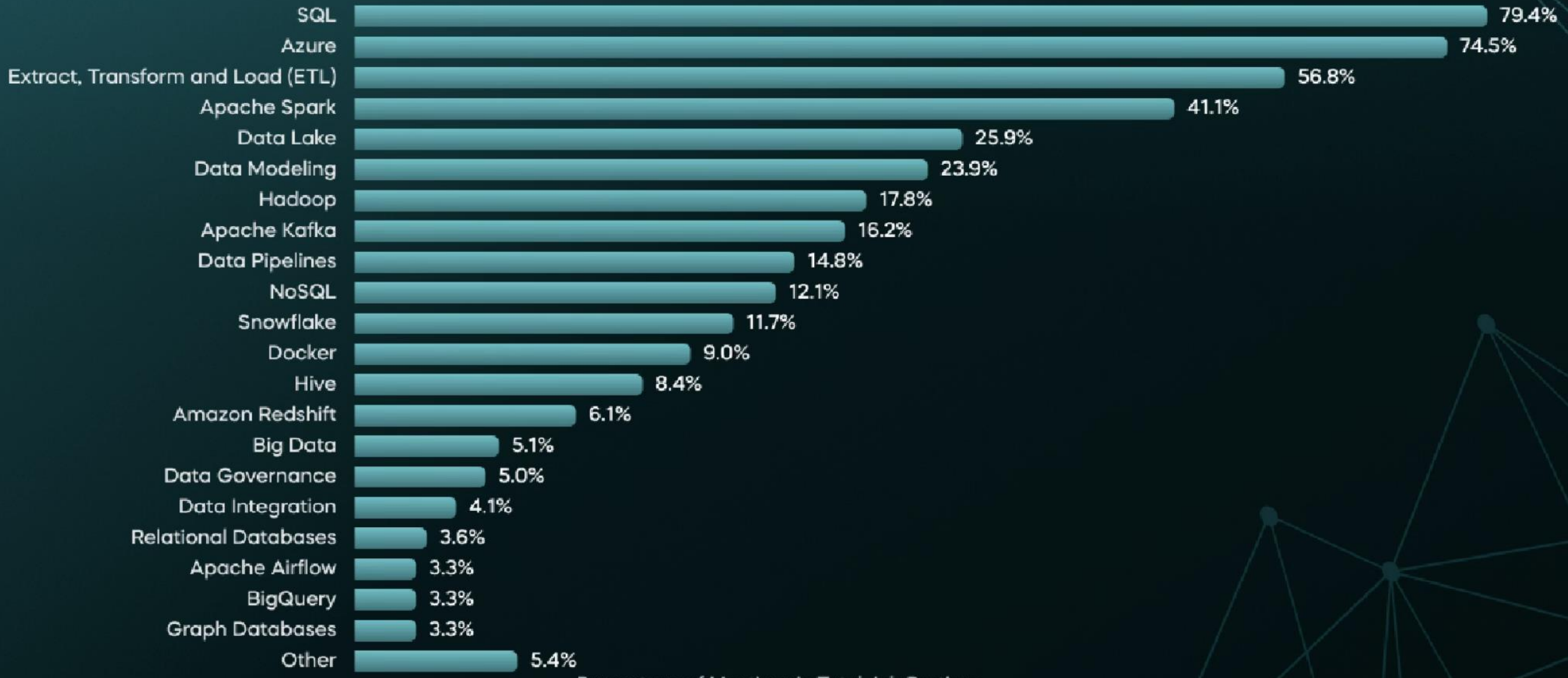
Sophie Magnet • 11 Sep 2024 • 18 min read

With the emergence of ChatGPT, recent years have seen concerns about how artificial intelligence (AI) could lead to job losses in data-related fields like data science and data engineering. There was speculation that AI could replace these roles because of its capacity to automate tasks traditionally performed by humans.

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365

Data Engineering Skills

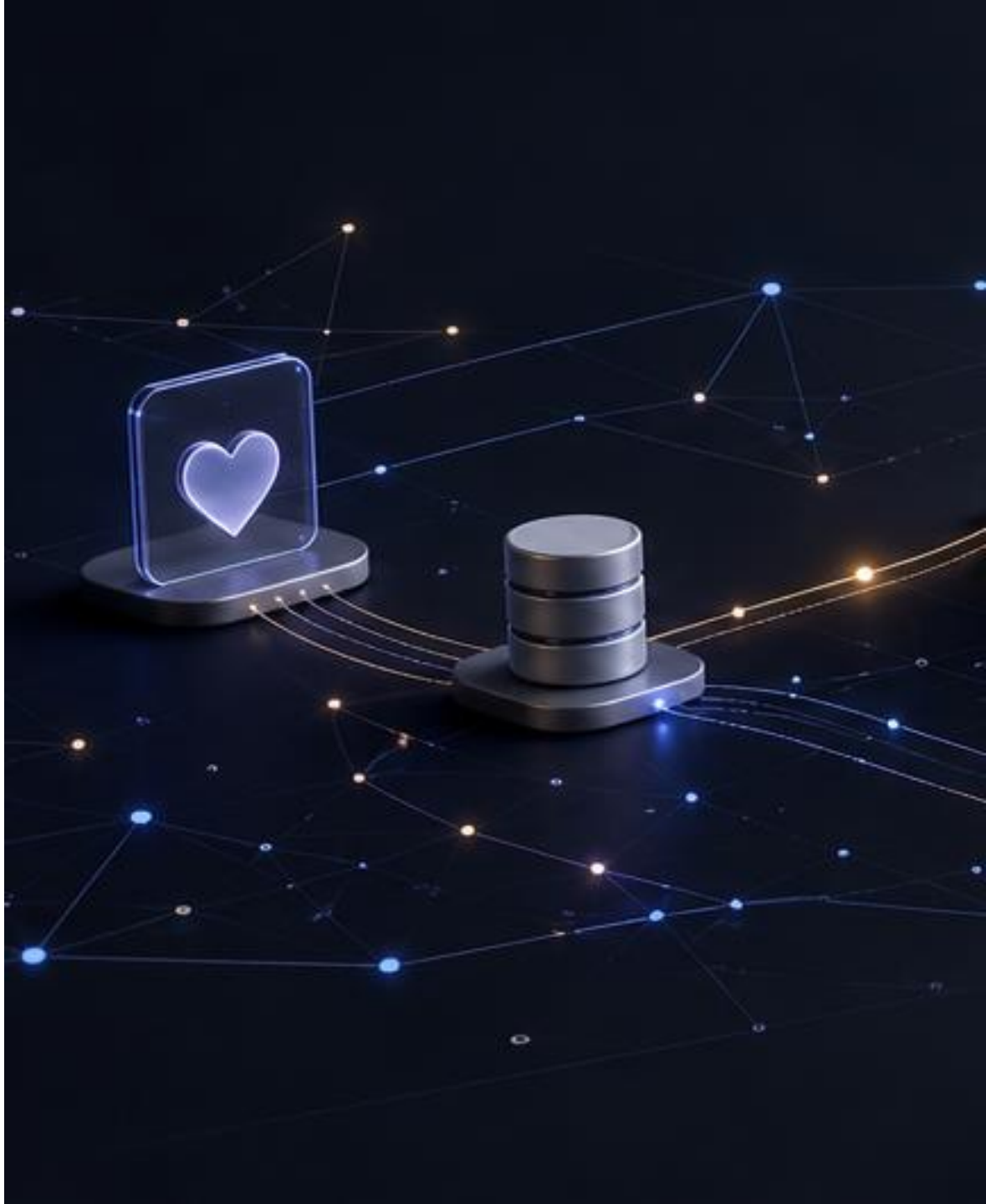


Data Engineering Skill Groups

- Database Management
- Big Data Technologies
- Data Warehouse / ETL
- Cloud
- Data Pipeline and Workflow Management
- Data Governance
- Containerization (e.g. docker, k8s)

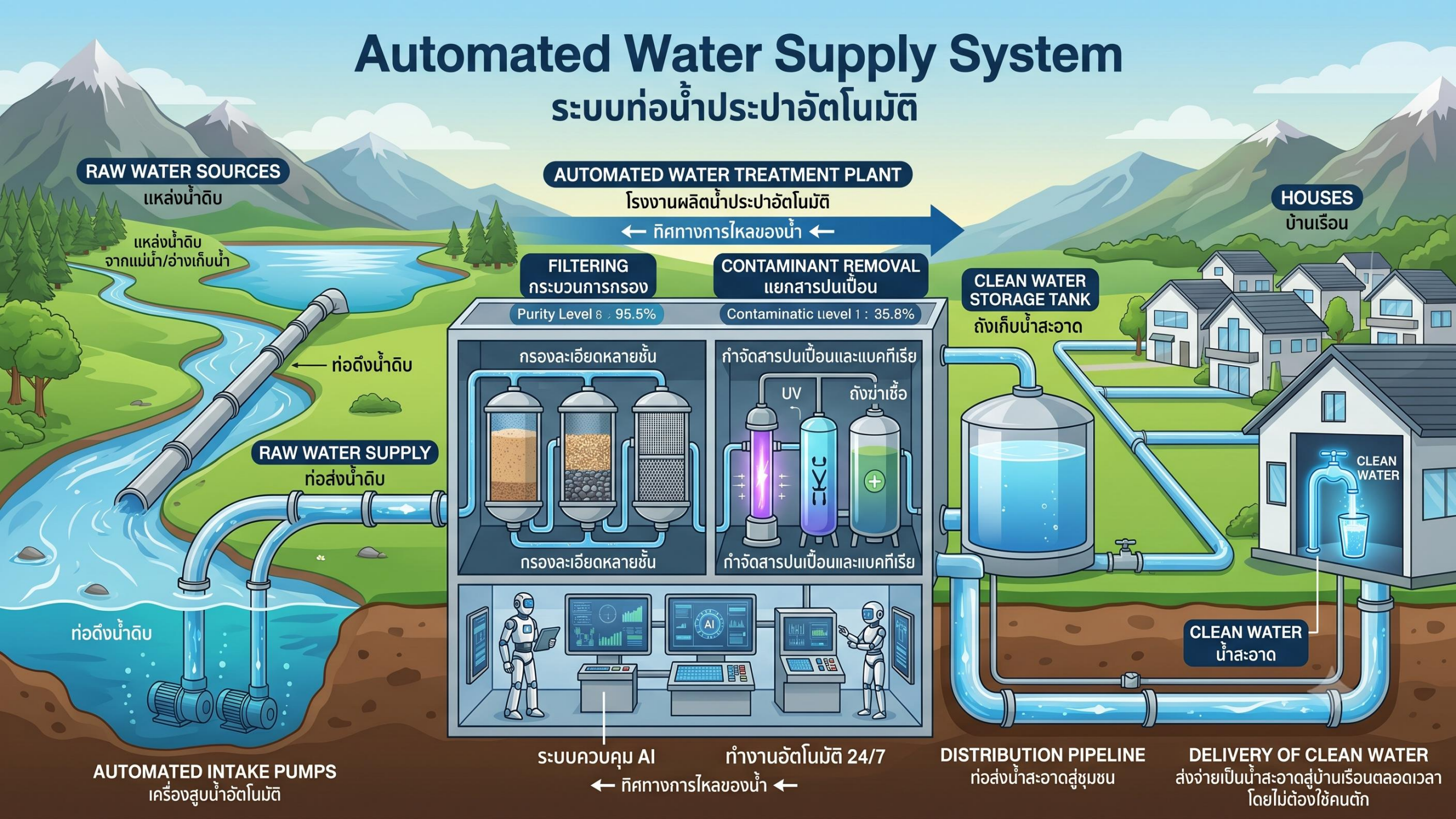
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Challenges in Data Engineering

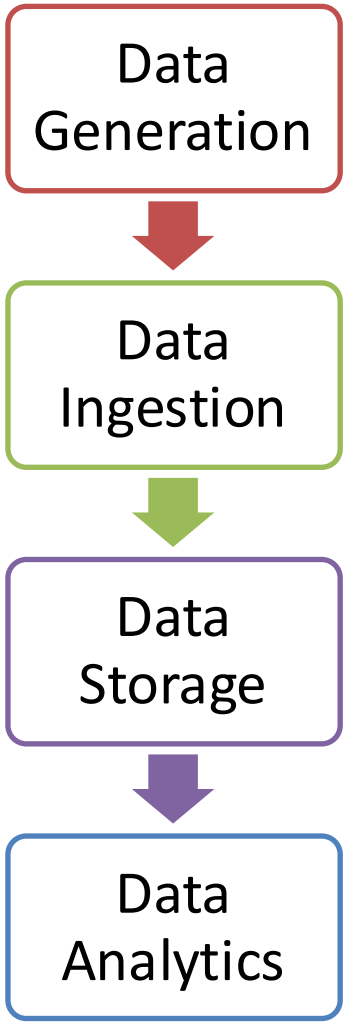


Automated Water Supply System

ระบบท่อน้ำประปาอัตโนมัติ



Anatomy of Data Pipeline



- Various data sources
- Various formats
- Real time and batch

- Transactional Databases
- Mobile Applications
- IOT / Sensors

- Data extraction
- Data transportation
- Data pre-processing

- Domain-specific protocol
- Middleware: Kafka, MQTT
- ETL tools

- Big data
- Relational vs NoSQL
- Data Lakehouse
- Query engine

- HDFS
- MongoDB, Cassandra, Redis
- Iceberg, Hudi, Ducklake
- Arrow, Presto, Drill, Trino

- Data processing
- Machine learning / deep learning
- Data visualization

- Pandas, Polars, Dask, Spark, Flink
- ML algorithms
- Power BI, Tableau, Metabase

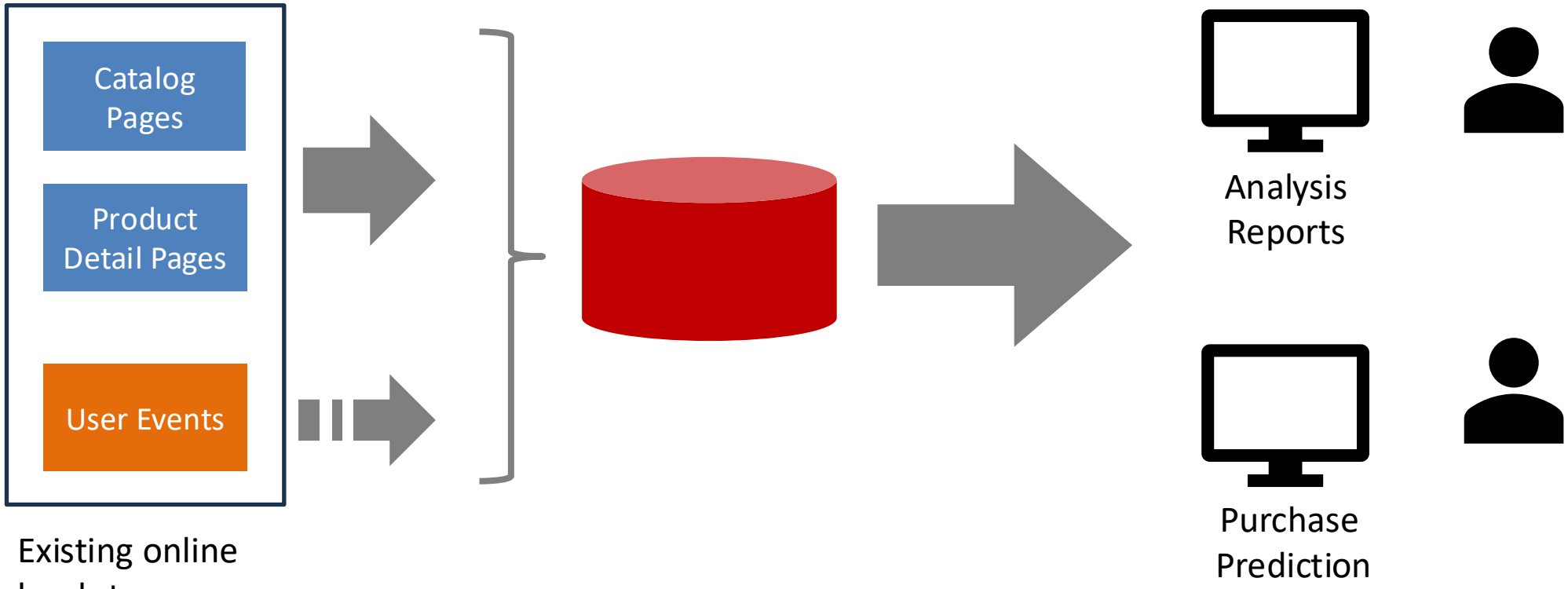
Single Source of Truth

- Important data must be correct, reliable, and have only one version
- To create single source of truth:
  - Data Centralization
  - Data Cleansing & Standardization
  - Master Data Management (MDM)
  - Data Governance and Data Catalog

BookPulse Scenario: One Website, Multiple Data Needs

- BookPulse inherits an online bookstore with no product API; product data exists only in HTML pages
- Catalog and product-detail pages contain different fields that must be combined into one product record
- Repeated scrapes must preserve raw pages and identify new products, removals, price changes, and stock changes
- Products are continuously updated; customers are still active and generate events, including views, cart additions, and purchases
- Analysts need reliable data for catalog reports, conversion analysis, session analysis, and purchase prediction

BookPulse Data Pipeline



Data pipeline handles information moving between systems

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BookPulse Data Pipeline Challenges

- Mix data source handling: Different data sources (structured / unstructured, real-time / batch) can be difficult to handle.
- Reliable data delivery: The right data must reach the intended system, recipient, or workflow without loss, duplication, or misrouting.
- Size and speed do matter: Handling large and fast data on a single server is mission impossible.

BookPulse Data Pipeline Challenges

- Timely synchronization: Data must be exchanged and updated within the timeframe required for safe, effective care
- Operational resilience: Errors must be detected quickly, contained effectively, and traced to their source for resolution and prevention

THE DATA SCIENCE HIERARCHY OF NEEDS

